ACM ShellSlayer Fall S026 Workshop

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ACM ShellSlayer Fall 2026 Workshop

Preamble

Visit: https://go.umd.edu/shellslayer

- Includes link to github page for ShellSlayer
- Includes a link to the **Cheat Sheet** in case you are stuck
- Has a Feedback Form to fill in at the end of this session

If you don't have Docker installed, punch "Install Docker" into a browser and install it!

Welcome Slayers

Immediate Goals

- Illustrate Docker as a way to get a shell environment / run programs
- Learn a some useful Unix commands and shell tricks that may improve your life on the command line
- Have some fun!

Long Term Goals

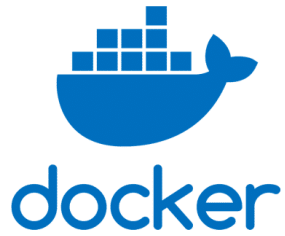
- Consider contributing to the tutorial via playtesting, writing levels, etc.
- Add some levels on some topics not covered

Thank yous

- UMD ACM for Inspiring the initial version of Shell Slayer
- All of you for choosing to show up!

Docker Setup

If you've not installed Docker already, search the internet for **"Install Docker"** and follow the instructions



- Docker is a system by which "containers" can be set up to run programs
- Containers provide a skeleton operating system along with libraries/utilities so that programs can run reliably
- Similar in spirit to Virtual Machines or Emulation BUT has some technical distinctions which you can read about on your own time
- Docker allows for download/run of containers with programs in them via a GUI (blegh) or command line (yay!)

Running Shell Slayer

Start the "Docker Desktop" application (via clicking things) then use a command shell to download / run the shellslayer image via:

Windows: `cmd.exe`

```
>> docker run -it --rm kauffman77/shellslayer
```

MacOS: `Terminal.app`

```
>> docker run -it --rm --platform=linux/amd64 kauffman77/shellslayer
```

Linux: Terminal Emulator of Choice

```
>> sudo docker run -it --rm kauffman77/shellslayer
```

sudo may not be necessary depending on your system config

Updating your Image

If you've previously done Shell Slayer, update your Docker image via

```
>> docker pull kauffman77/shellslayer
```

Structure of the Exercise

daemon (noun)
1. Archaic spelling of demon
2. In multitasking computer operating systems, a program that runs as a background process rather than being under the direct control of an interactive user.

You sense a foul presence in the process tree...

- A "Daemon" process is present and must be killed.
- To do so, you must switch to be the Root user (administrator)
- Becoming root requires a password
- Each level (`level_00_less`, `level_02_editing`, ...) teaches about a Unix tool/technique and when completed properly will reveal part of the root password (one of the Daemon's names)
- Complete all levels, get the whole password, and slay the Daemon

Workshop Plan

Level	Name	Overview
`level_00_less`	1	Viewing text files with `cat` / `less`
`level_01_touch`	-	`touch` to create /freshen files
`level_02_editing`	2	Editing text files with `vi` / `nano`
`level_05_grep`	3	Searching for text with `grep`
`level_99_su_kill`	-	Changing to "root" and killing processes
UNUSED		Omitted for time
`level_10_sh_loops`	4	Looping over files in the shell
`level_15_find`	5	Locating files with `find`
`level_20_sed`	6	Substituting text in files with `sed`

For each level...

- Give a short overview
- Give participants a few minutes to try commands, solve the exercise
- Regroup to discuss the stage solution and fill in a Daemon name

Extended Version

The UNUSED levels can be run using

```
>> docker run -it --rm kauffman77/shellslayer /main.sh extend
```

for those that want to study the topics in them

Thank You for Your Participation

Participant Feedback at `https://go.umd.edu/shellslayer`

- If you enjoyed the workshop and have feedback, provide it in the survey
- If you are interested in contributing to the tutorial, email me
  - Chris Kaufman
  - UMD Computer Science
  - `<profk@umd.edu>`



The world doesn't need a hero.
It needs a professional.